

Prospects and Facets of Quantum Magic

Raghav G. Jha

Based on arXiv: 2606.09971 + upcoming work

Magic and entanglement in 1+1-dimensional SU(2) lattice gauge theory

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Entanglement and non-stabilizerness (magic) quantify two distinct departures of quantum systems from classical description: the former measures non-local correlations, while the latter measures the deviation from stabilizer states that can be efficiently simulated classically. Understanding magic in physically relevant quantum field theories is essential for identifying where quantum advantage may be realized in the early fault-tolerant quantum computing era. We calculate the gauge-invariant entanglement entropy and stabilizer Rényi entropy of the ground state of the (1+1)-dimensional SU(2) lattice gauge theory formulated in a dressed-site basis that enforces Gauss's law exactly. Using tensor networks, we obtain results for system sizes up to $L = 100$ (300 qubits). We find a crossover denoted by g_* where the ground state passes from a more magic-rich regime into a regime with less magic; this is also tracked by the sharpest change of both the entanglement entropy and lattice particle density. Our large-scale study of non-stabilizerness and entanglement entropy in a non-Abelian lattice gauge theory with matter provides new insight into the interplay of magic and entanglement in gauge theories, both of which are relevant for classical and early fault-tolerant quantum simulations.

In review @ PRL

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[rgjha.github.io](https://github.com/rgjha)

Outline

- How do we define hardness? What is magic? Why is it important?
- Why are we interested in studying lattice gauge theories?
- How do we truncate the $SU(2)$ gauge theory with matter in $1+1$ -dimensions keeping only “physical” (i.e., gauge singlet) states?
- How do we measure magic and entanglement entropy for these LGTs?
- Results + ongoing work

Clifford group and GK theorem

- $\{H, S, CX\}$ are generators of the Clifford group which is a normaliser of the Pauli group i.e.,
 $\mathbf{C}_n = \{V \in U_{2^n} \mid V\mathbf{P}_n V^\dagger = \mathbf{P}_n\}$.
- Gottesman-Knill (GK) theorem says that any quantum computation — 1) where the qubits are prepared in the computational basis, 2) acted on by Clifford gates, 3) and measured in the computational basis can be simulated efficiently on a classical computer.

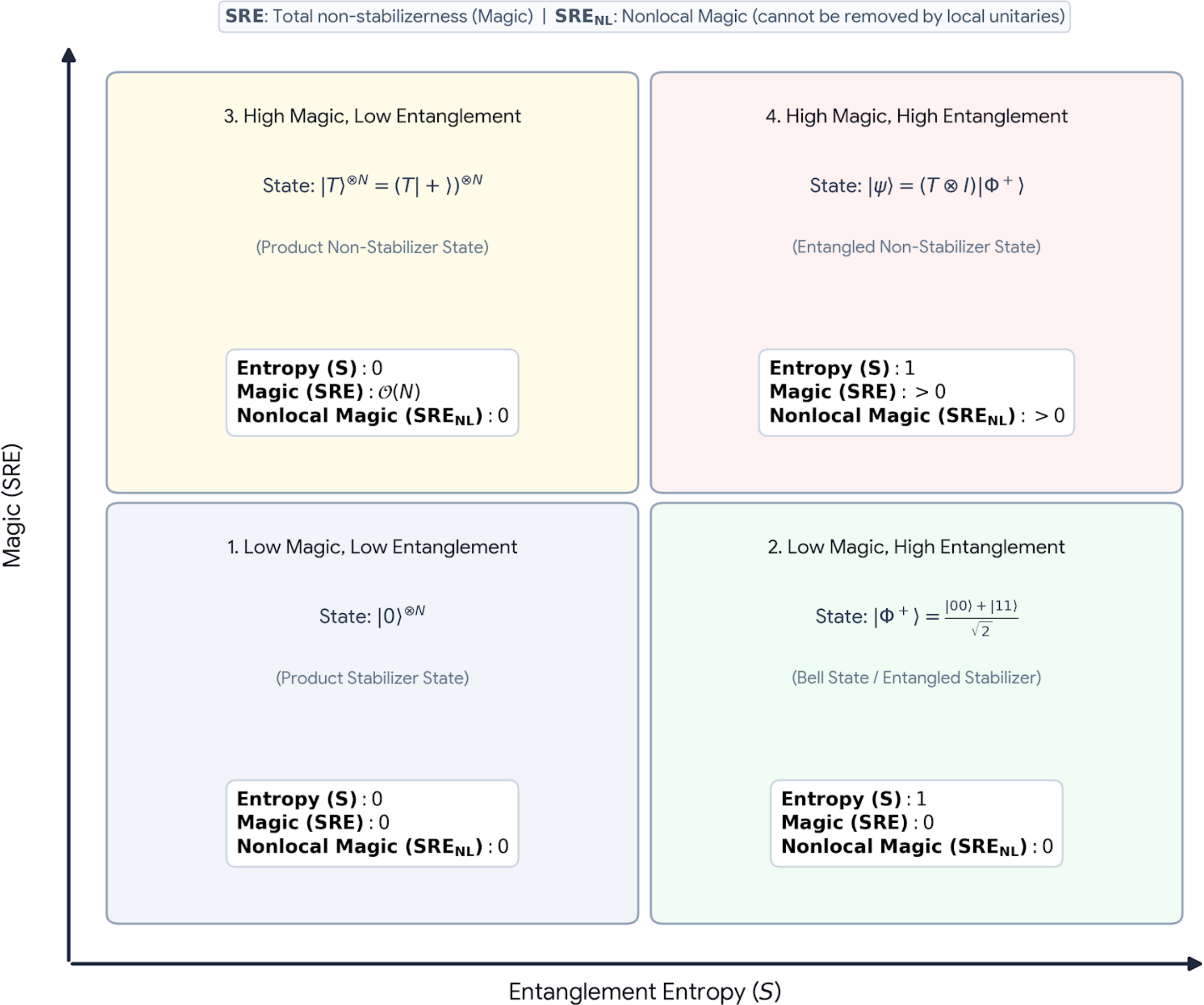
Example: Bell state

- $(|00\rangle + |11\rangle)/\sqrt{2}$ i.e., maximally entangled state of two qubits. Can be prepared from $|00\rangle$ by acting with H and CX gate.
- Even though it is maximally entangled, it is a stabilizer state (any state reachable by Clifford gates only starting with state in computational basis). Using GK theorem, we know that this can be efficiently (polynomial) simulated on classical computers.
- We thought that more entanglement leads to classical non-simulability. This is not true! This is stabilizer entanglement
- There must be some other different/observable that captures this classical non-simulability and makes quantum computers powerful.

Magic (or non-stabilizerness)

- We ought to look for non-stabilizerness as providing the departure of what can and what cannot be simulated efficiently. This has been termed as `magic`.
- In very *simple* terms, more magic means harder for classical computers.
- Also needed for any universal quantum gate set. For example, Clifford + T-gate is universal since T-gate supplies the magic!
- Connecting magic to an observable that can be efficiently computed is not straightforward.
- In recent years, based on work of [*Leone et al.*](#), we have a measure of magic in terms of what is called `Stabilizer Renyi Entropy' (SRE).
- SRE is zero for product and stabilizer states, while it is non-zero for say $T^{\otimes n} |0\rangle^{\otimes n}$.

Resource plane

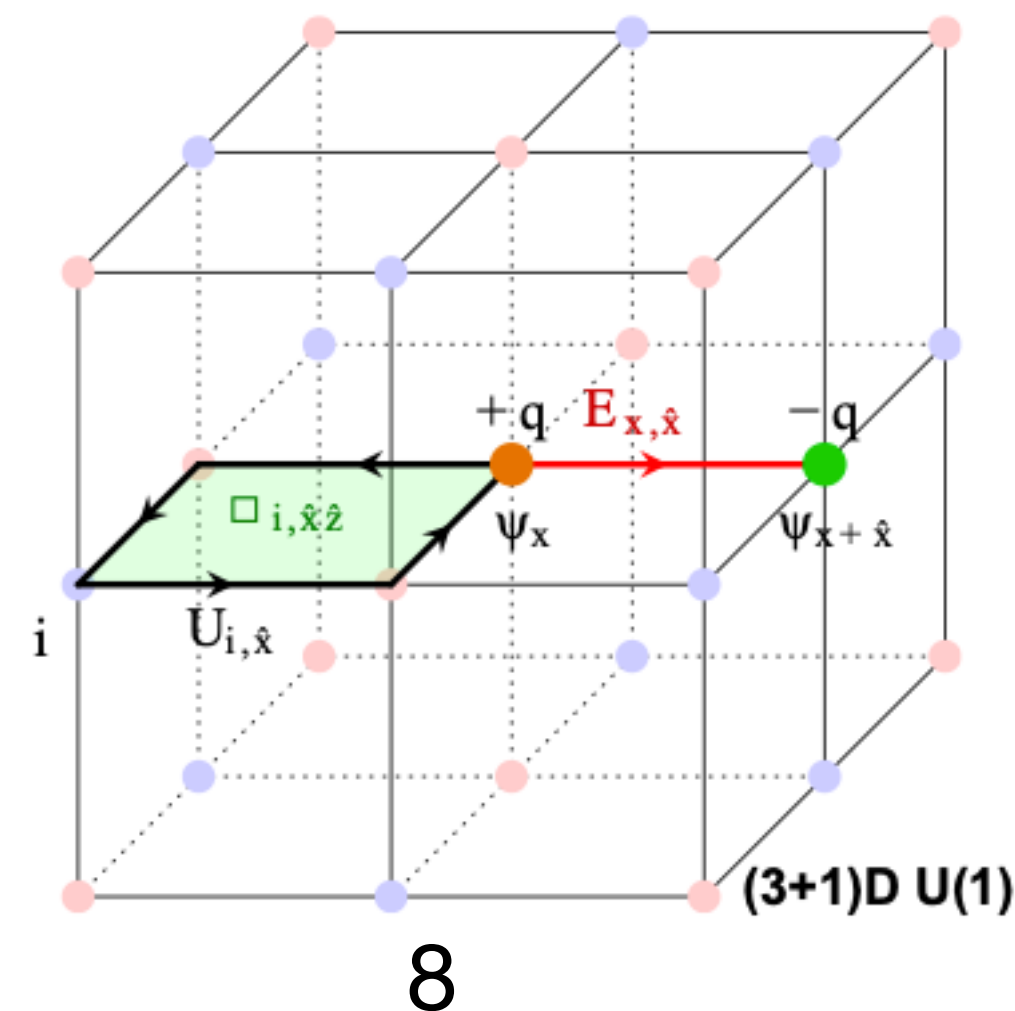


Magic as an observable

- Though, magic first appeared in the context of universal quantum computation around 2001-2005, we did not had a nice mathematical way to quantify it.
- In 2021, Leone et al., proposed a definition (to be defined later) where we can compute amount of magic for a state by computing the expectation values over all possible Pauli strings of the system. If the state is an eigenstate, or some state that can be generated by Clifford group, this quantity is zero.
- This has led to ease of computation since Pauli strings are natural objects in a digital quantum computation. However, computing this observable is exponentially hard. Some success has been achieved using MCMC sampling.

General lattice gauge theory

- In order to understand quantum field theory, one works with lattice gauge theory and then obtains the continuum limit which described the QFT one wanted to initially study. This approach was pioneered by Wilson (in 1974, Nobel Prize Physics, 1982).
- In most of the numerical work till date, the lattice gauge theory (LGT) has been considered in the Lagrangian approach (by defining Euclidean action). On the other hand, Hamiltonian lattice gauge theories have not been so extensively studied. This is partly due to lack of tools in hand using this approach.
- Kogut-Susskind gave the first prescription of what Hamiltonian LGTs should look like. In general, we have four terms known as electric, magnetic, mass, and hopping (gauge-invariant term) that connects matter and gauge fields.



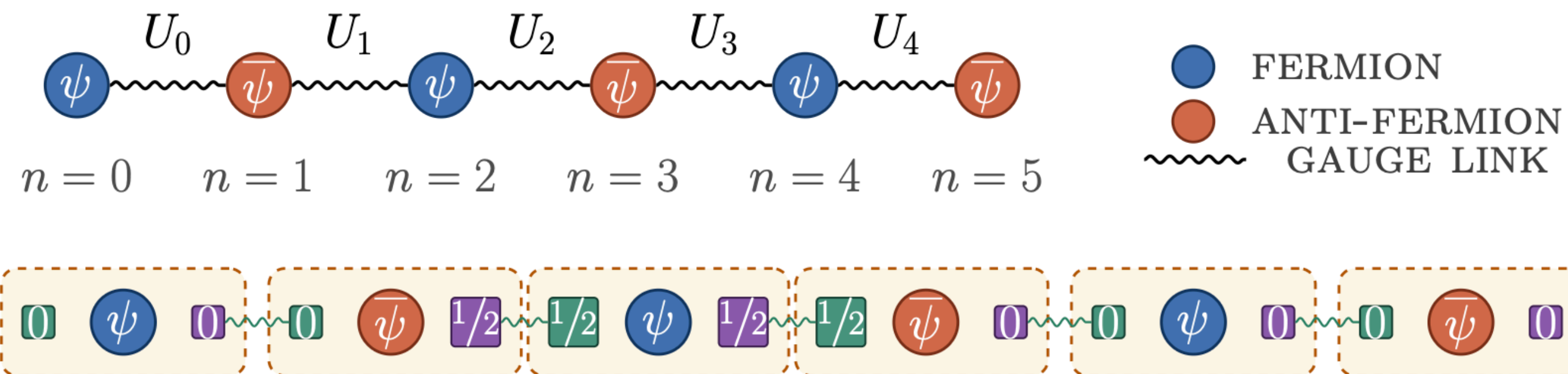
Kemper et al, arXiv: 2511.13882

1+1 SU(2) gauge theory

- We consider simplest example of non-Abelian gauge theory in 1+1-dimensions. There are different approaches for truncating the infinite-dimensional Hilbert space. We follow the dressed-site/electric basis approach where at each lattice site, we collect the semi- gauge links on the left and right of the site into a dressed site. Then the physical lattice with matter and gauge fields restricted to physical gauge-singlet states are dressed sites connected by Abelian conservation law.
- We restrict to $j_{\max} = 1/2$, extensions to higher $j_{\max}$ can be done. We did not consider this, however provide some details about construction of Hamiltonian and gauge-invariant space in the Appendix of the paper.

1+1 SU(2) gauge theory

- The setup is as follows:



- We restrict to $j_{\max} = 1/2$, so each link (or semi-link) can either values 0, $1/2$. The matter fields (at physical sites) are represented as $\psi, \bar{\psi}$ in the standard staggered fashion. The vacuum and doubly occupied matter sites are singlets, while we have one doublet.

1+1 SU(2) gauge theory

- $|0\rangle = |j, m = 0\rangle$, and $|d\rangle = (\psi_n^r)^\dagger (\psi_n^g)^\dagger |0\rangle$ are singlets while $(\psi_n^r)^\dagger |0\rangle$, $(\psi_n^g)^\dagger |0\rangle$ are doublets. In short, for matter sites, we have $\mathbf{1} \oplus \mathbf{2} \oplus \mathbf{1}$. For semi-links, we have $\mathbf{1} \oplus \mathbf{2}$ for $j_{\max} = 1/2$.
- When we combine these in a dressed site which is gauge-singlet, we have to find singlets in following expansion: $(\mathbf{1} \oplus \mathbf{2}) \otimes (\mathbf{1} \oplus \mathbf{2} \oplus \mathbf{1}) \otimes (\mathbf{1} \oplus \mathbf{2})$ which is 36-dimensional local space.
- One can show that there are 6 gauge singlets, i.e., 6/36 states are physical! We therefore build a Hamiltonian acting on these states of size $6^L \times 6^L$, where L is the number of lattice sites. We need $3L$ qubits.

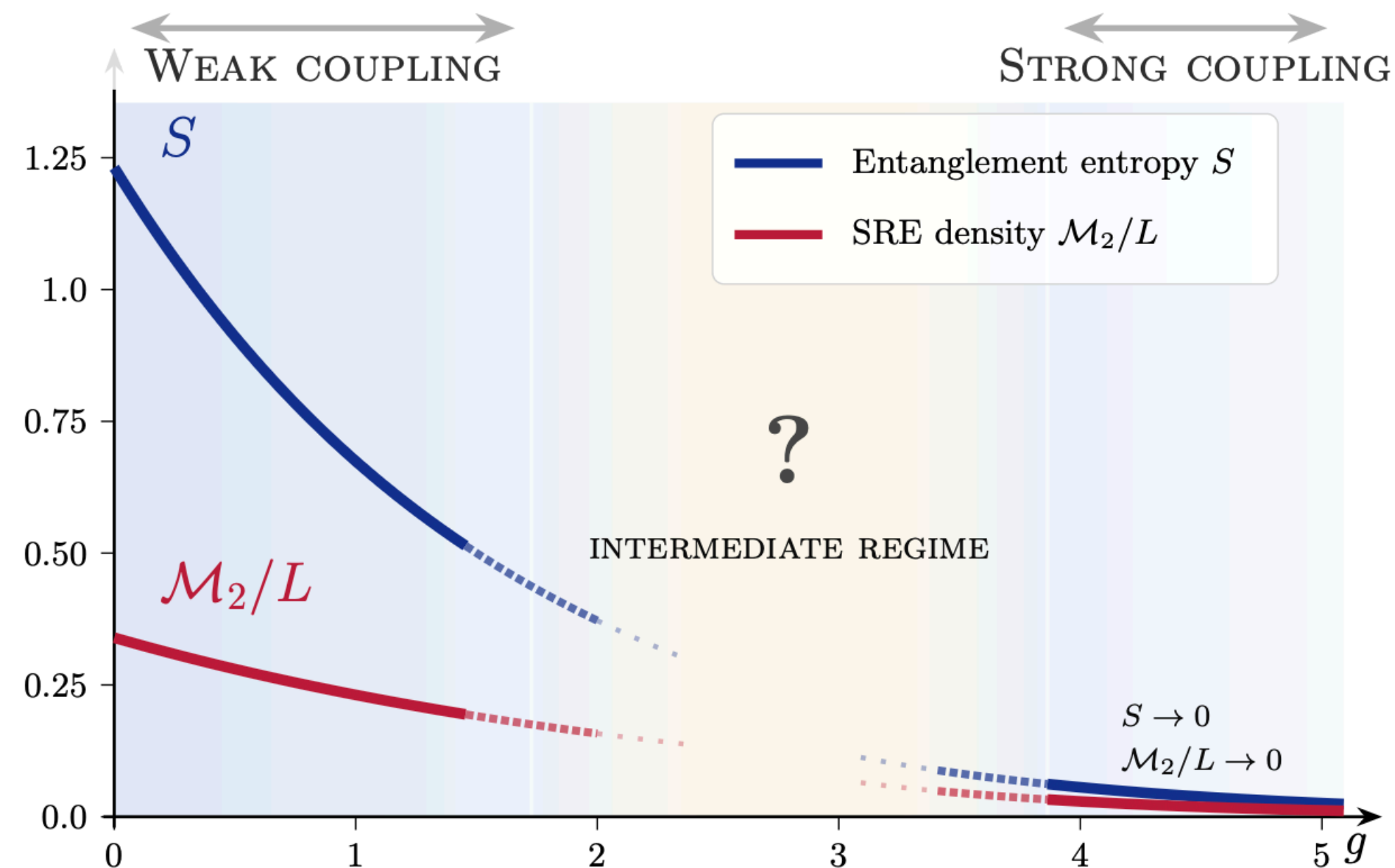
1+1 SU(2) gauge theory

- The Hamiltonian is given by: $\sum_n \left[g^2 \hat{C}_n + m (-1)^n \hat{M}_n + \hat{A}_n^1 \hat{B}_{n+1}^1 + \hat{A}_n^2 \hat{B}_{n+1}^2 \right]$, where the operators $\hat{C}, \hat{M}$ are diagonal and the hopping operators, $\hat{A}, \hat{B}$ raise or lower the flux by 1/2. More details in the paper.
- We will probe entanglement entropy (EE) and magic in the $m - g$ plane. Note that these are dimensionless lattice parameters converted from dimensionful parameters by using powers of a .

1+1 SU(2) gauge theory - expectations!

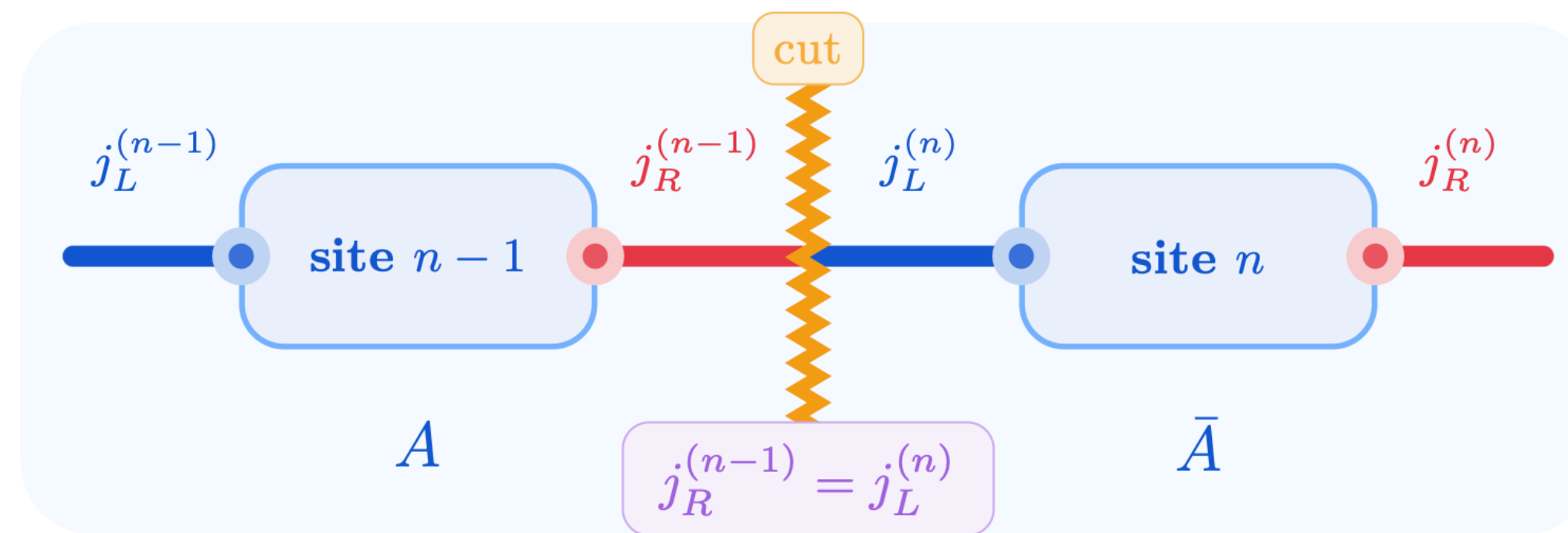
- Following Dirac, *we do not start a sentence without knowing its end*. How should the results look like?
- CFTs are magical and possess high magic [White *et al.*, [arXiv: 2007.01303](#)] so we should find maximum magic around $g \rightarrow 0$. In that limit, we also have high entanglement (Calabrese-Cardy, 2004). As $g \rightarrow \infty$, we know from the structure of ground state that it should have zero magic (product state) and no entanglement (Donnelly, 2011).

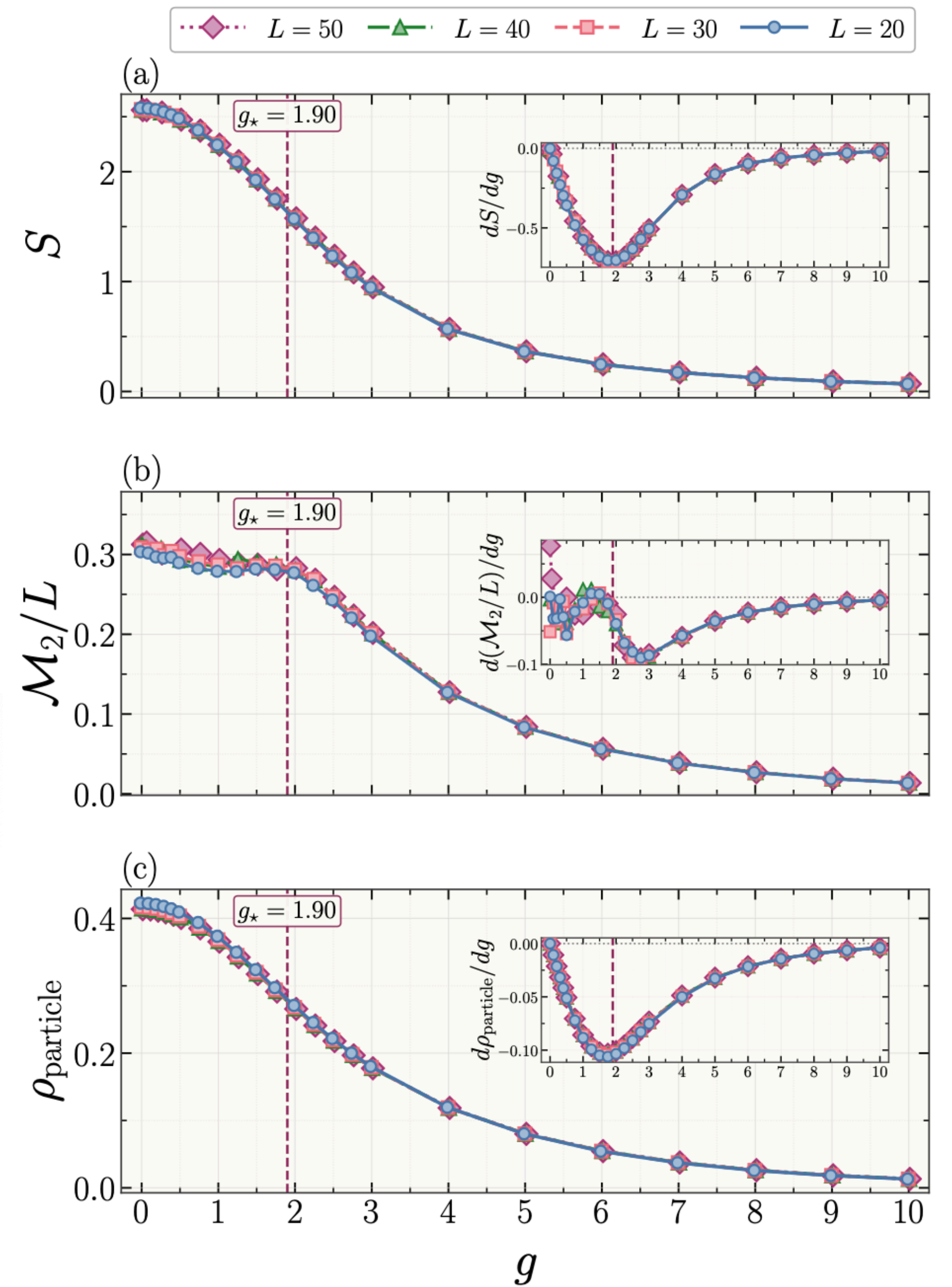
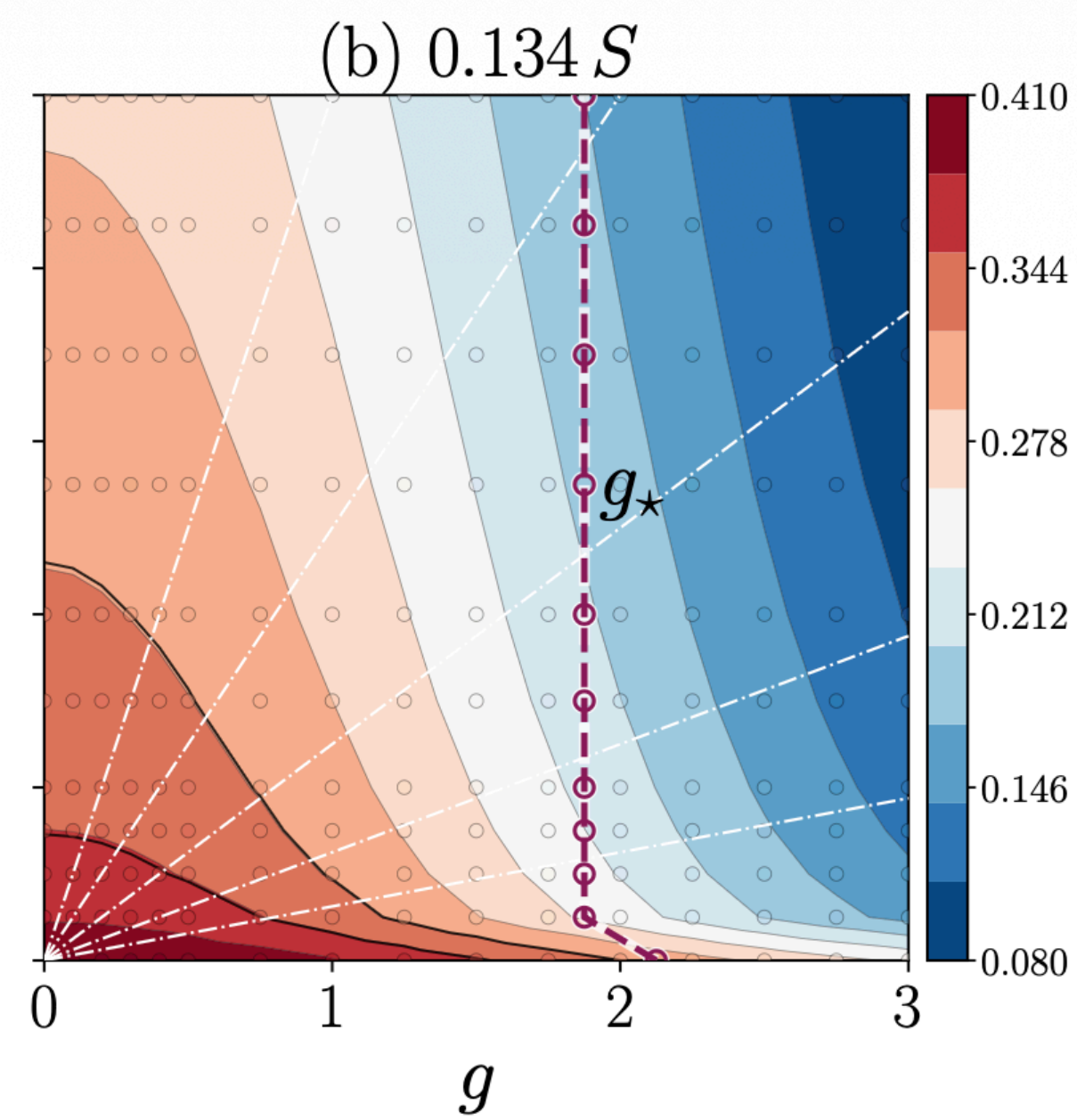
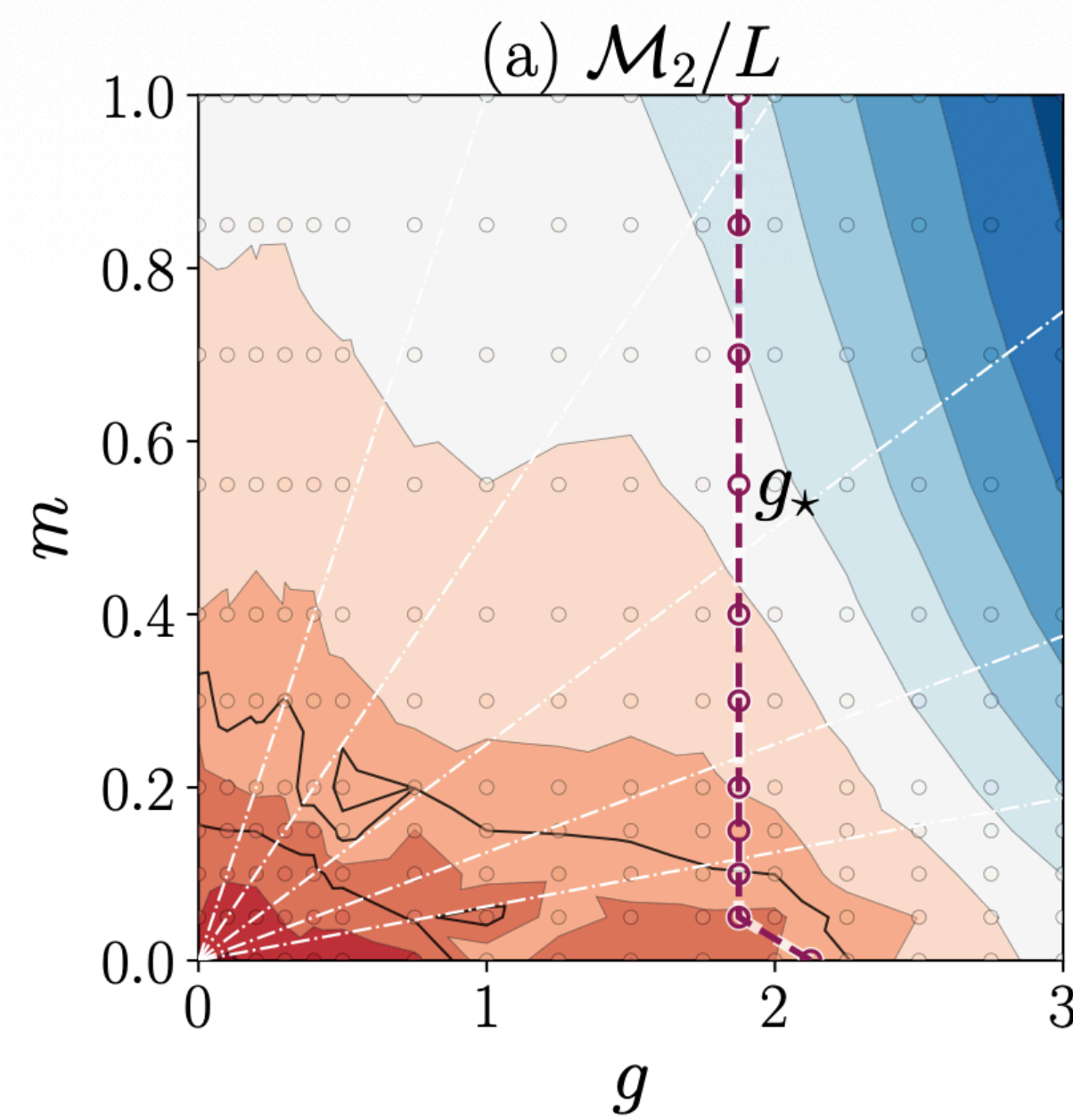
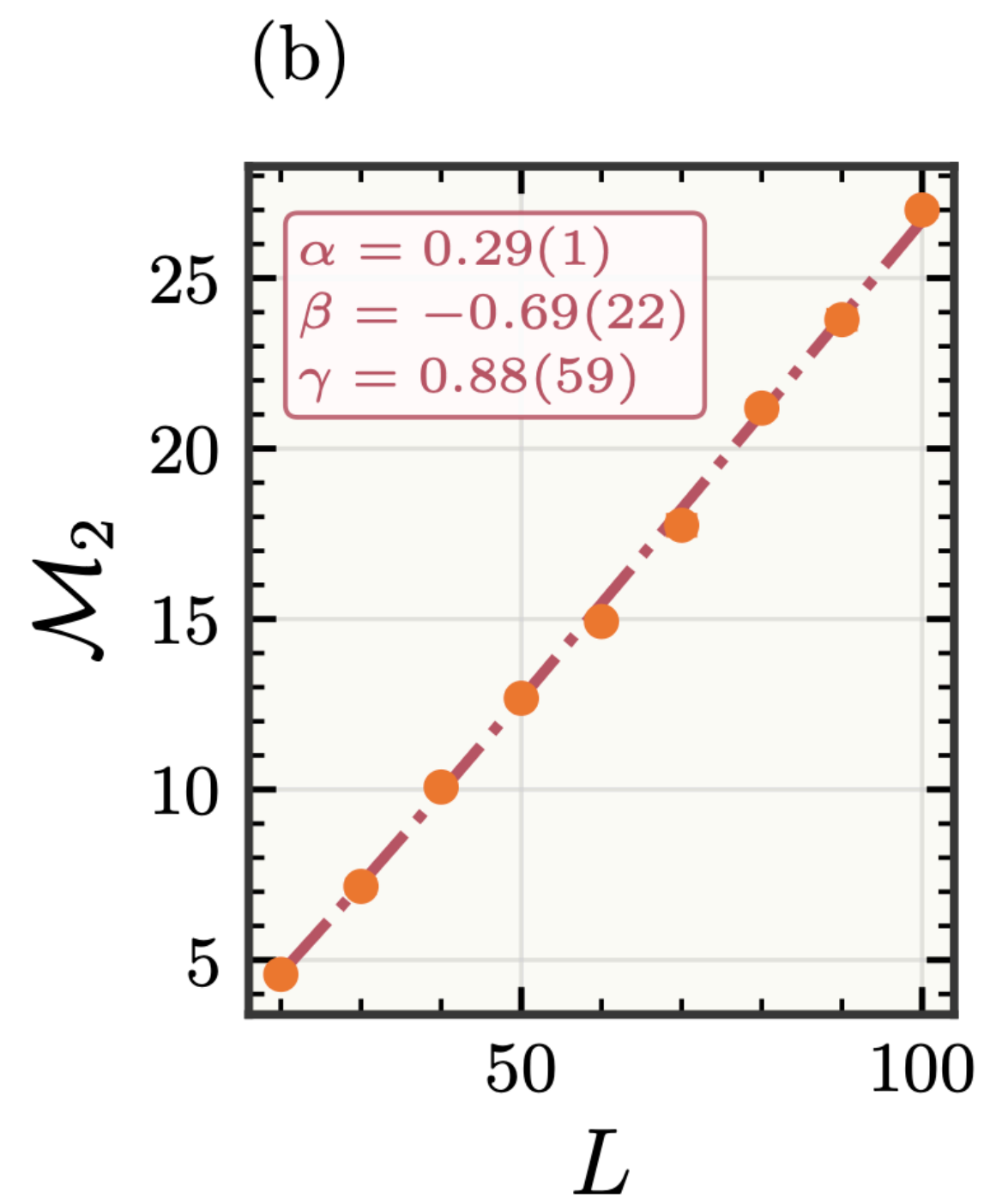
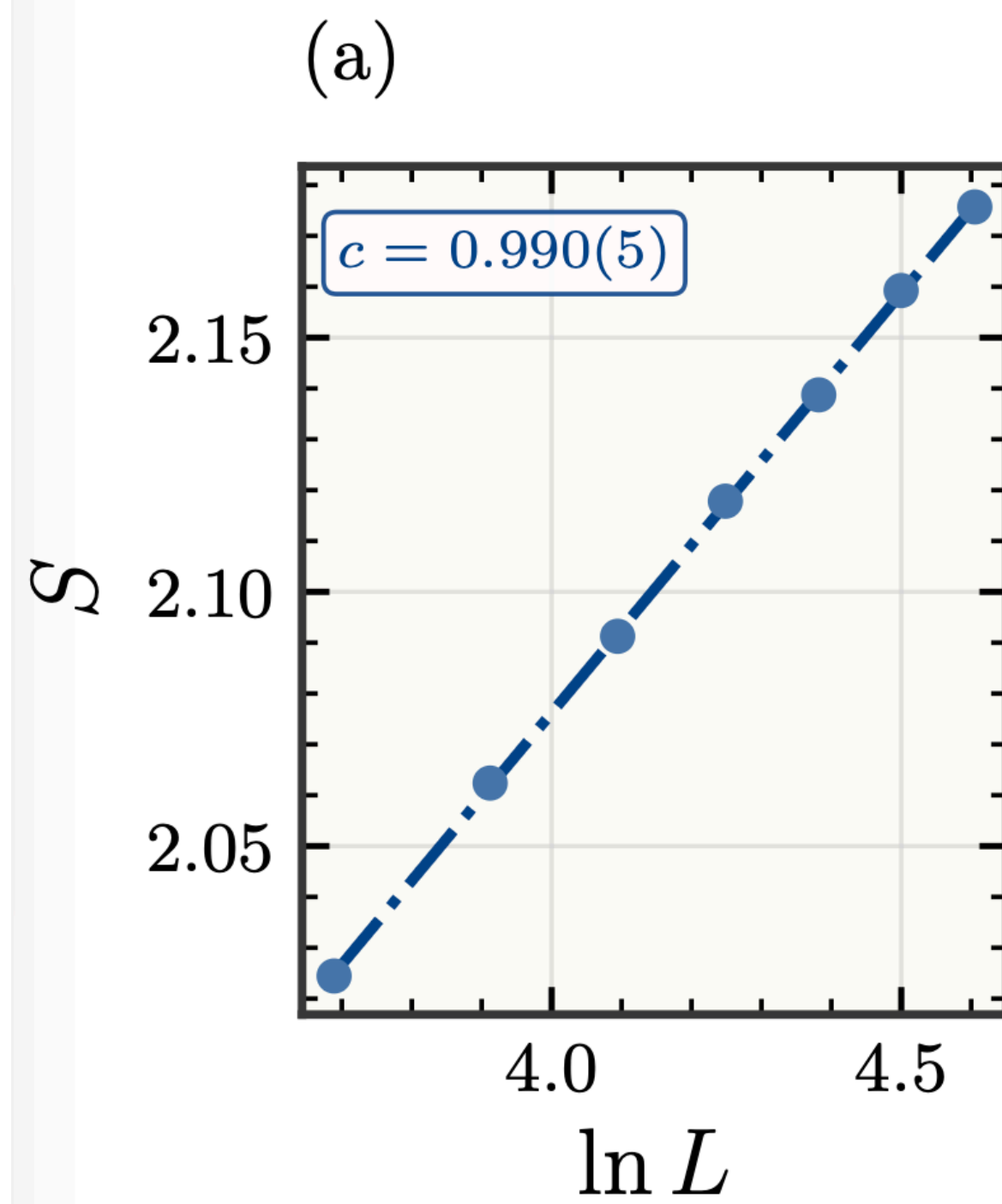
- So, we have:



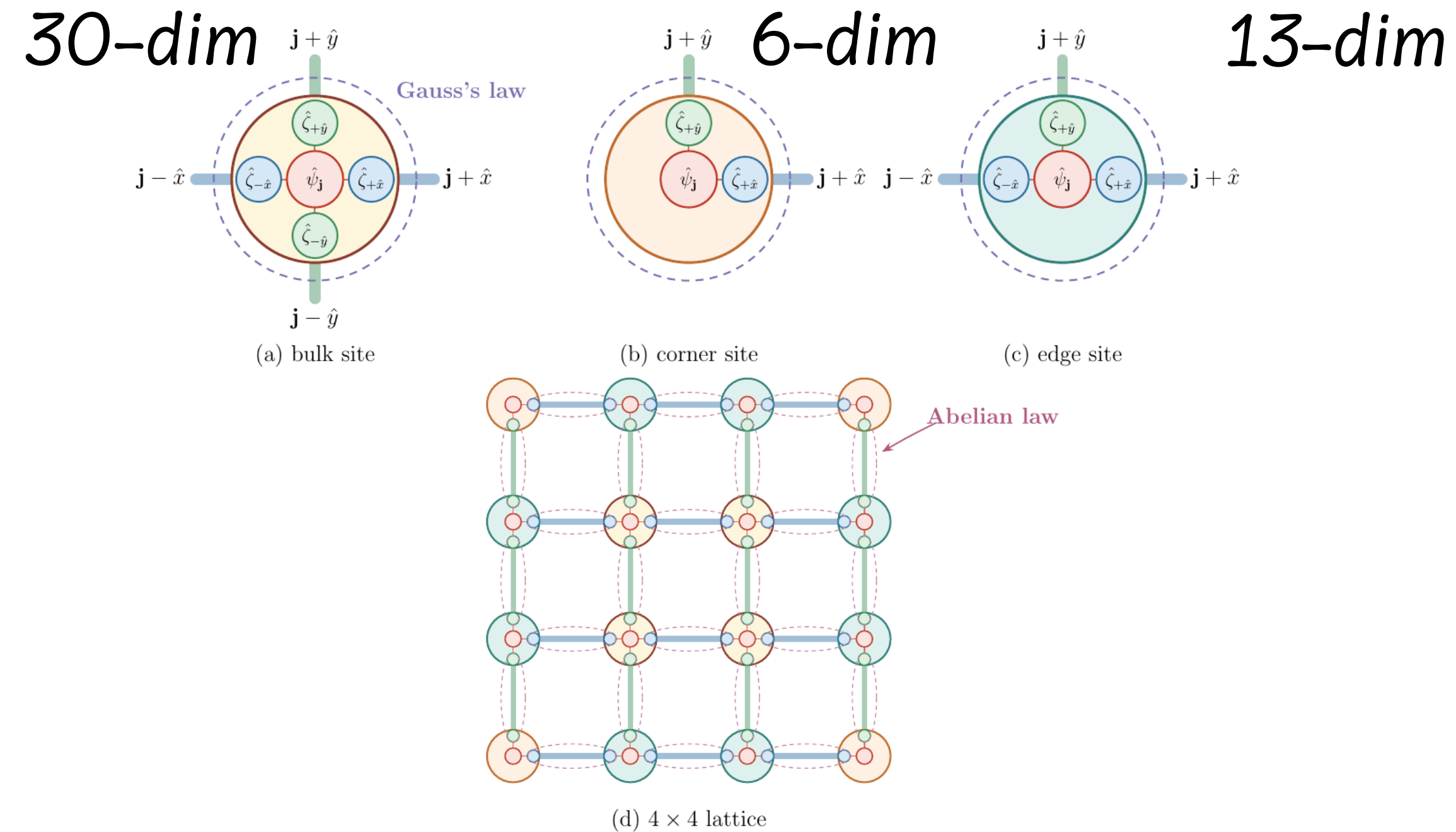
EE and SRE

- For magic, we compute the stabilizer Renyi entropy (SRE) defined as: $M_n = \frac{1}{1-n} \log_2 \left(\frac{1}{2^N} \sum_{P \in \mathcal{P}_N} |\langle P \rangle|^{2n} \right)$, where n is the order of the SRE (we consider $n=2$), N is the total number of qubits of the system. Computing this is exponentially hard but many MCMC based methods have been developed which can help us compute these for large system sizes.
- For entanglement entropy, we use the extended-Hilbert space formulation [Donnelly, 2011]. EE can then be written as sum of three terms. See our paper for more details.





2+1 SU(2) gauge theory



- The setup is as follows:

- We restrict to $j_{\max} = 1/2$ again, so each link (or semi-link) can either values 0, 1/2. The matter fields (at physical sites) are represented as $\psi, \bar{\psi}$ in the standard staggered fashion. The number of gauge singlets is related to a geometry problem of counting number of non-intersecting chords that can be drawn between n points on the circle — Motzkin numbers (related to Catalan number).

2+1 SU(2) gauge theory

- The Hamiltonian is the standard Kogut-Susskind type Hamiltonian with electric, magnetic, mass, and hopping terms.

$$\begin{aligned}
 H = \frac{1}{2a} \sum_{j \in \Lambda} \sum_{\alpha, \beta} & \left[-i \psi_{j, \alpha}^\dagger U_{j, j+\mu_x}^{\alpha\beta} \psi_{j+\mu_x, \beta} \right. \\
 & \left. - \epsilon_j \psi_{j, \alpha}^\dagger U_{j, j+\mu_y}^{\alpha\beta} \psi_{j+\mu_y, \beta} + \text{H.c.} \right] \\
 & + m_0 \sum_j \epsilon_j \sum_{\alpha} \psi_{j, \alpha}^\dagger \psi_{j, \alpha} + \hat{H}_{\text{pure}}, \quad (15)
 \end{aligned}$$

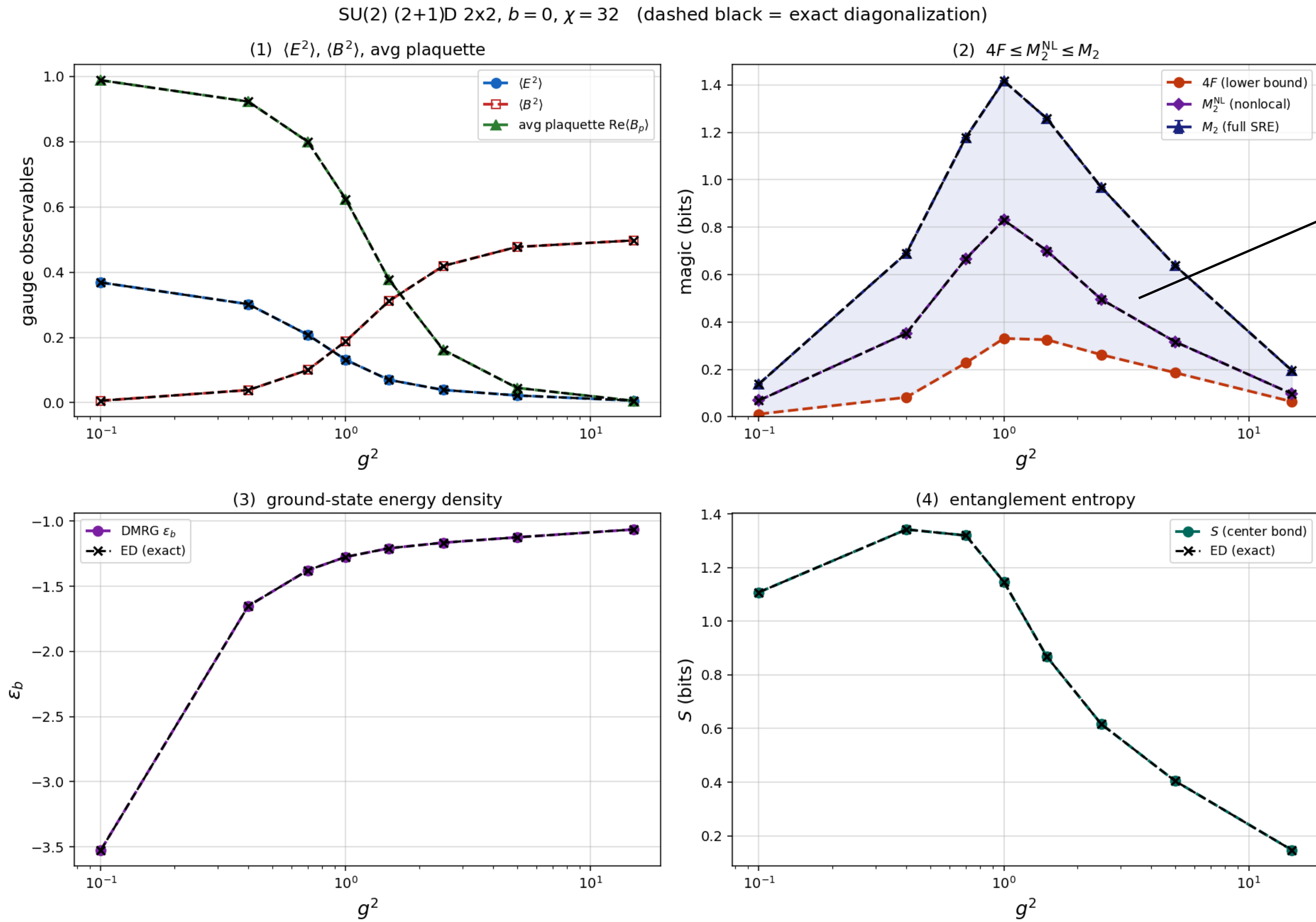
where

$$\hat{H}_{\text{pure}} = \frac{g^2}{2a} \sum_j \left(\hat{E}_{j, j+\mu_x}^2 + \hat{E}_{j, j+\mu_y}^2 \right) - \frac{8}{ag^2} \sum_{\square} \text{Re Tr } \square, \quad (16)$$

where $\square = U_{j, j+\mu_x} U_{j+\mu_x, j+\mu_x+\mu_y} U_{j+\mu_y, j+\mu_x+\mu_y}^\dagger U_{j, j+\mu_y}^\dagger$
and we have set $c = \hbar = 1$.

- $H_E \sim g^2, H_M \sim 1/g^2$ and therefore as we take small values of g , the magnetic term becomes more relevant. There is usually a crossover/transition due to competition between these terms.

2+1 SU(2) gauge theory - Preliminary results



What are these?

2+1 SU(2) gauge theory - Preliminary results

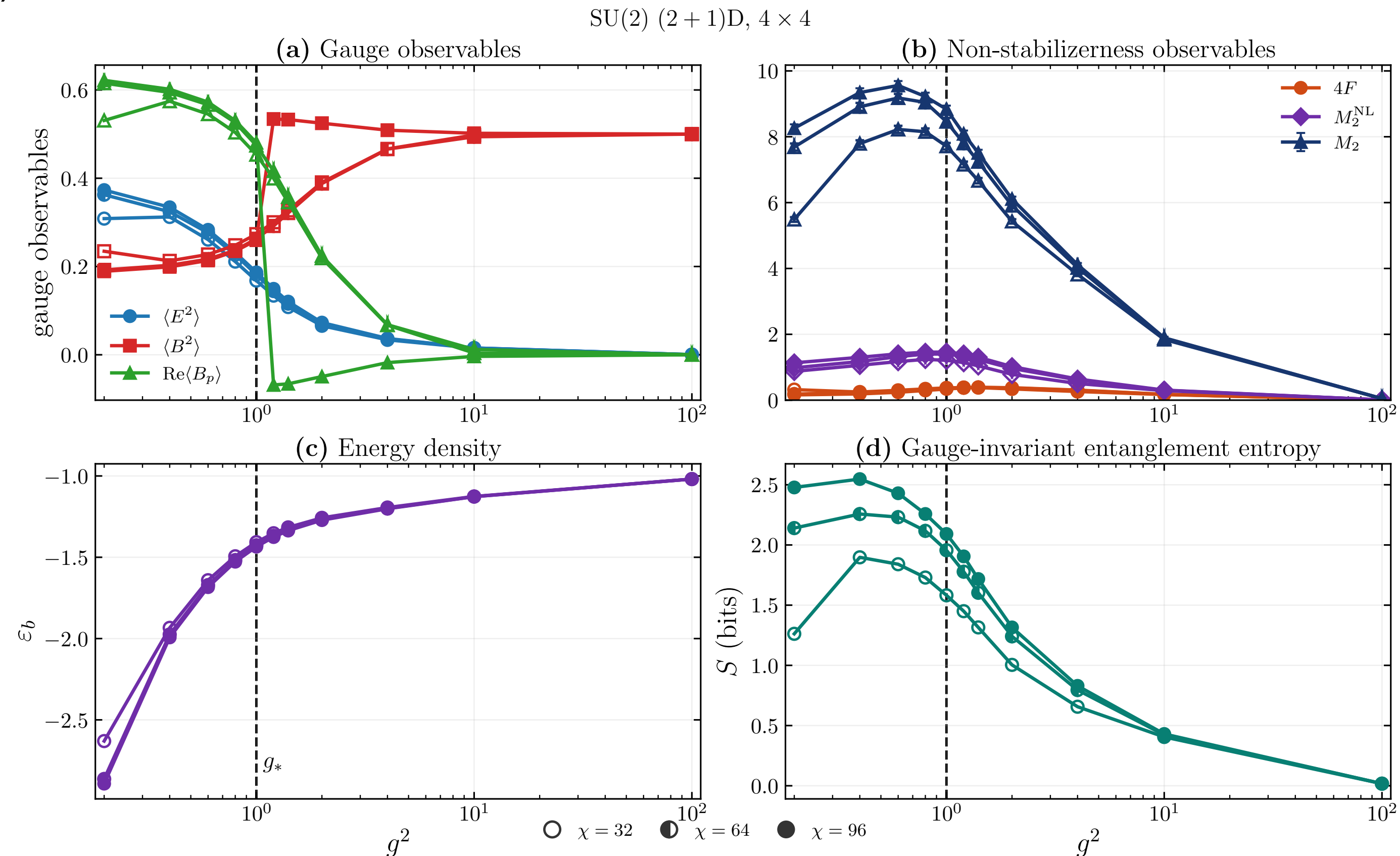
- Usually, we compute SRE as explained earlier. But, this observable is extensive and is the full magic. Recently, it has been realised (*Cao et al, arXiv: 2403.07056*) that there is a component of this SRE which is more important for understanding the true quantum correlations. This is referred to as non-local magic!
- This magic is defined (for some bipartition) like the entanglement entropy, and is mathematically defined as: $M_2^{\text{NL}}(|\psi\rangle) = \min_{U_A \otimes U_B} M_2((U_A \otimes U_B) |\psi\rangle)$. Here, A and B are the subsystems into which the state has been split.
- For Schwinger model and other spin models, it was found that this has critical signatures and can detect onset of phase transitions. Has not been explored much for non-Abelian gauge theories in general.

2+1 SU(2) gauge theory - Preliminary results

- In fact, it is also known that non-local magic is lower bounded by something known as 'anti-flatness' (AF).
- AF measures how flat/similar the entanglement spectrum actually is. For example, for Bell state after tracing out one of the qubits, the spectrum is flat.
- It is defined as: $F_n(\rho) = \text{Tr}\rho^{n+1} - (\text{Tr}\rho^2)^{\frac{n-1}{n}} = \sum_i \lambda_i^{2n+2} - \left(\sum_i \lambda_i^{2n}\right)^{\frac{n-1}{n}}$, usually studied for $n = 2$
where $\rho = \rho_A = \text{Tr}_B(\rho)$.

2+1 SU(2) gauge theory - Preliminary results

- Using tensor network simulations, we can scale the lattice size. We find peak in non-local magic around the g^2 that was reported in a previous paper from computing non-magic observables (*Cataldi et al, arXiv: 2307.09396*)



Future directions

Time evolution of magic

- We have focused on the static (ground state) properties of magic till now. However, one of the main questions, for quantum advantage related to dynamics, is how does the magic grows with time evolution.
- In 1d, tensor networks are powerful tool to simulate quantum dynamics. Is there a way we can quantify the growth of magic and find a regime where quantum advantage might lie for certain class of Hamiltonians (and initial states)?

Following entanglement with magic!

- The entanglement entropy ideas entered physics in the 1990s, when people were exploring CFTs and quantum gravity. It is now well-established and forms the basis of the success of classical tensor network methods.
- Around October 2005, Levin-Wen and Kitaev-Preskill found part of entanglement entropy which is topological. They showed that ordinary entanglement entropy contains a universal constant that directly detects long-range topological order, even though topological order has no ordinary local order parameter.

Topological entanglement entropy

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We formulate a universal characterization of the many-particle quantum entanglement in the ground state of a topologically ordered two-dimensional medium with a mass gap. We consider a disk in the plane, with a smooth boundary of length L , large compared to the correlation length. In the ground state, by tracing out all degrees of freedom in the exterior of the disk, we obtain a marginal density operator ρ for the degrees of freedom in the interior. The von Neumann entropy $S(\rho)$ of this density operator, a measure of the entanglement of the interior and exterior variables, has the form $S(\rho) = \alpha L - \gamma + \dots$, where the ellipsis represents terms that vanish in the limit $L \rightarrow \infty$. The coefficient α , arising from short wavelength modes localized near the boundary, is nonuniversal and ultraviolet divergent, but $-\gamma$ is a universal additive constant characterizing a global feature of the entanglement in the ground state. Using topological quantum field theory methods, we derive a formula for γ in terms of properties of the superselection sectors of the medium.

PACS numbers: 03.65.Ud, 71.10.Pm, 73.43.Nq

arXiv: hep-th/0510092

Detecting topological order in a ground state wave function

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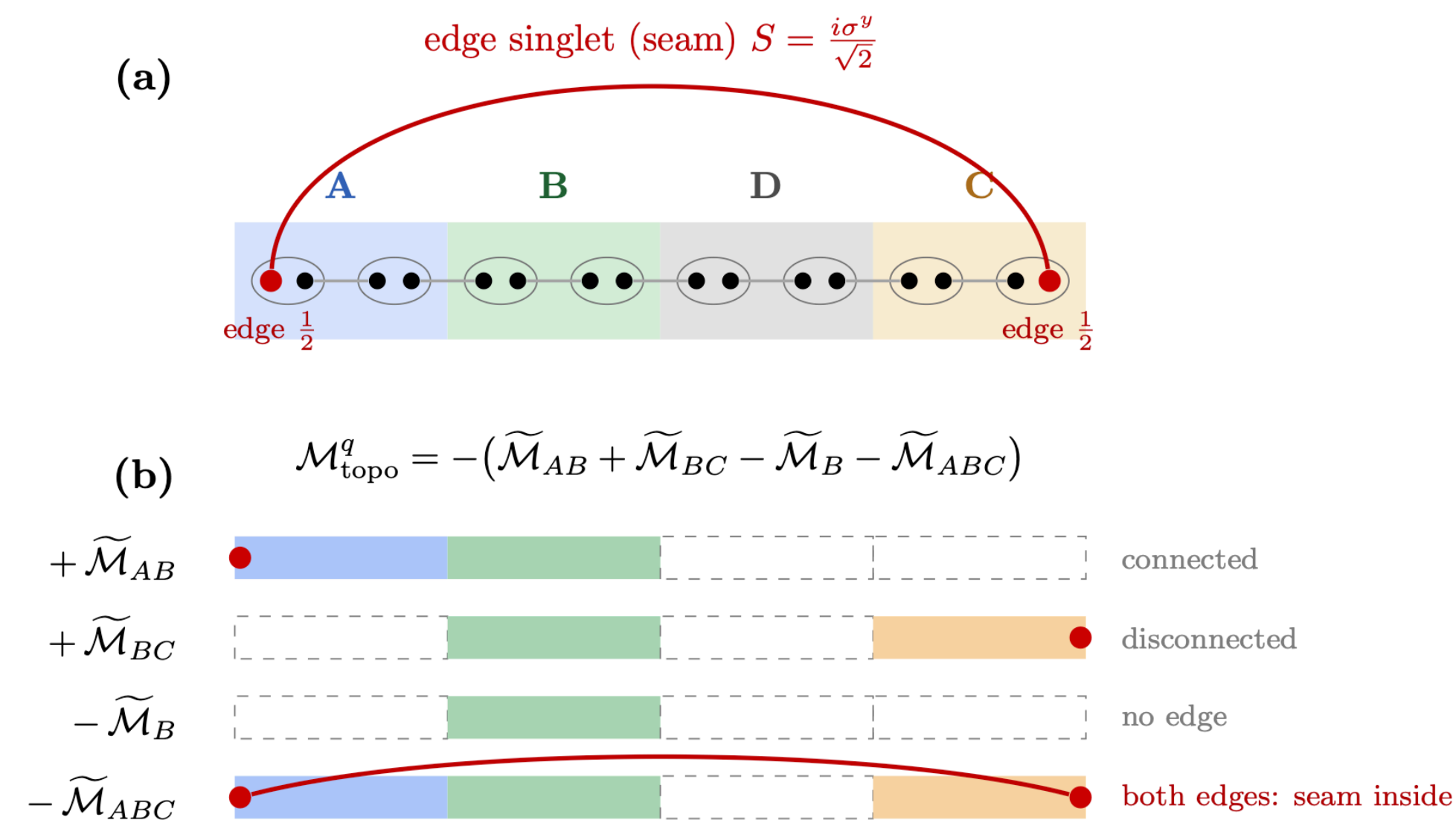
A large class of topological orders can be understood and classified using the string-net condensation picture. These topological orders can be characterized by a set of data $(N, d_i, F_{lmn}^{ijk}, \delta_{ijk})$. We describe a way to detect this kind of topological order using only the ground state wave function. The method involves computing a quantity called the “topological entropy” which directly measures the quantum dimension $D = \sum_i d_i^2$.

PACS numbers: 11.15.-q, 71.10.-w

cond-mat/0510613

Entanglement and Magic!

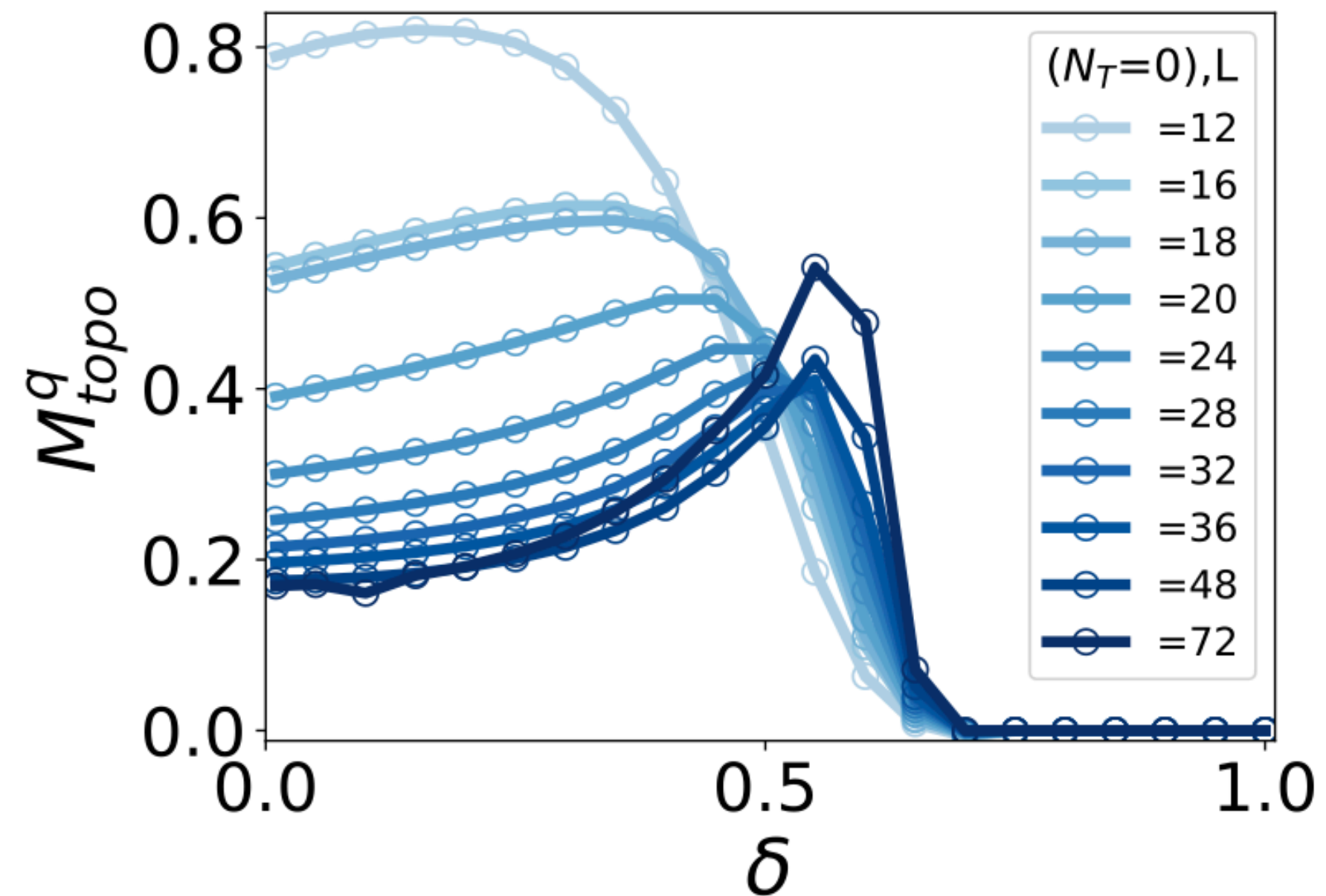
- Recently, ([Nehra et al, arXiv: 2512.16673](#)), proposed topological SRE (TSRE) that can be used to detect topological phases.
- TSRE is defined as: $M_q^{\text{topo}} = - \left(\widetilde{M}_{AB} + \widetilde{M}_{BC} - \widetilde{M}_B - \widetilde{M}_{ABC} \right)$. The designing of the traced over and retained parts is such that any bulk contribution is cancelled and exposes universal part.
- We are trying to understand TSRE and attempting to derive a closed form expression that agrees with numerical results presented in the above paper. For this, one useful place to start is the AKLT model since it can be exact described by an MPS with a very small bond dimension of $\chi = 2$.



Entanglement and Magic!

- The Hamiltonian is given by: $H_{\text{gAKLT}} = (1 - \delta) \sum_{\ell=1}^{L-1} \left[\mathbf{S}_{\ell} \cdot \mathbf{S}_{\ell+1} + \frac{1}{3} (\mathbf{S}_{\ell} \cdot \mathbf{S}_{\ell+1})^2 \right] + \frac{\delta}{L} \sum_{\ell=1}^L (S_{\ell}^z)^2$.

The TSRE for range of δ (numerically) was found out. In the Haldane phase ($\delta = 0$), it was found to go to 0.169. Is it possible to get this result analytically?



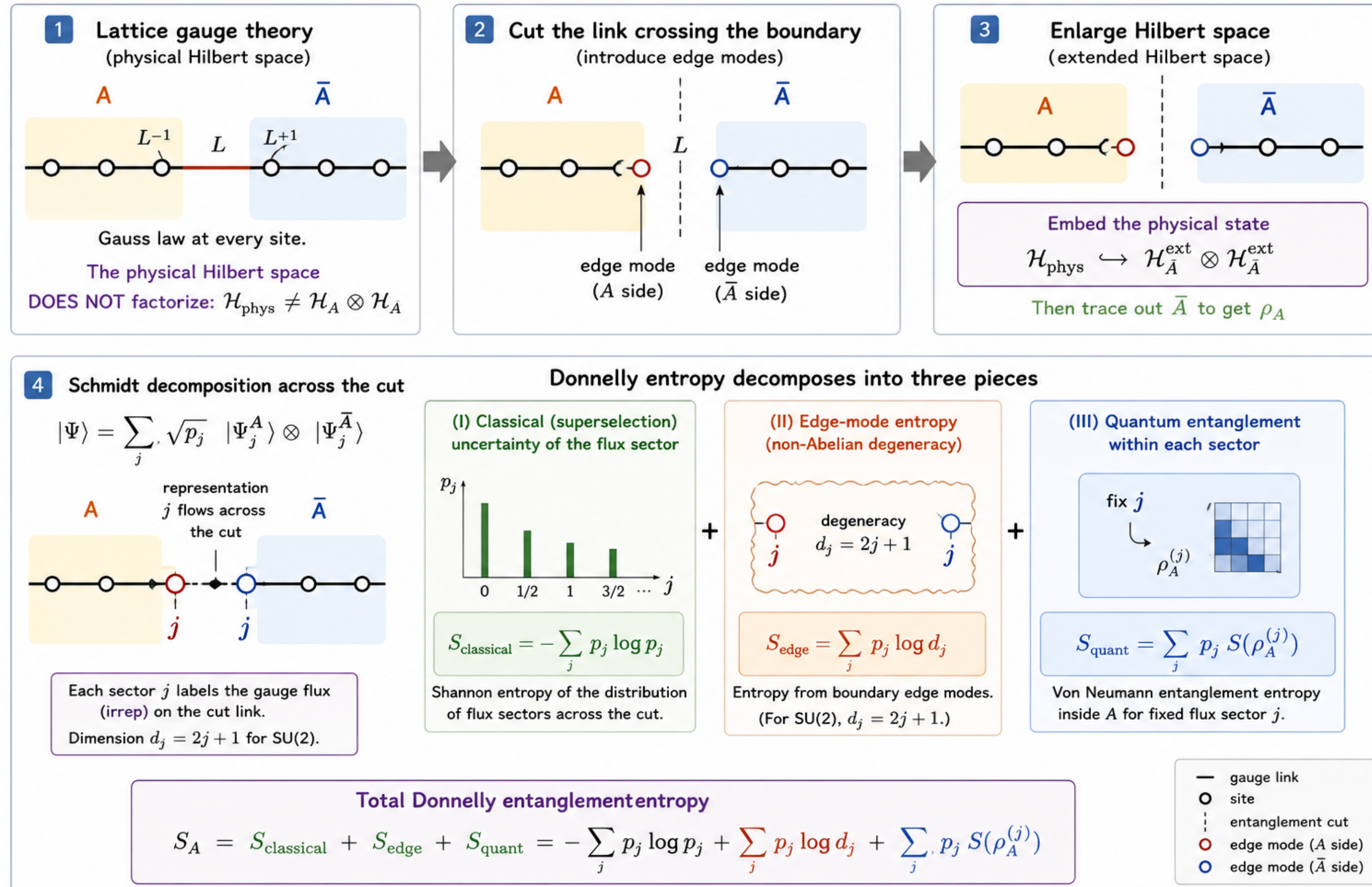
Summary

- We find that continuum limit of LGTs correspond to QFT states that have high magic. This has drastic implications for future quantum simulations of lattice gauge theories. It sort of implies that reaching continuum limit of Hamiltonian LGTs needs substantial (non-classical) quantum resources . One can think of this in terms of T-gates required.
- In this truncated 1+1d, SU(2), magic and entanglement need not necessarily track each other (i.e., they act as different quantum resource measure) since while EE can be *consistently* decreasing, magic can stay constant just before weak-strong crossover.
- In 2+1-dimensional SU(2), we found that non-local magic is a useful probe for phase transitions and can sometimes reveal quantum correlations/QPT earlier (or for smaller bond dimension) than EE. Full implications are not fully understood.
- In addition, understanding TSRE in models such as SSH model, and its connection to Zak/Berry's phase is also work in progress.

Thank you

Backup

Donnelly prescription and entropy decomposition



Backup

$$|1\rangle = |0, 0, 0\rangle,$$

$$|2\rangle = \frac{|\textcolor{red}{r}, 0, \textcolor{green}{g}\rangle - |\textcolor{green}{g}, 0, \textcolor{red}{r}\rangle}{\sqrt{2}},$$

$$|3\rangle = \frac{|\textcolor{green}{g}, \textcolor{red}{r}, 0\rangle - |\textcolor{red}{r}, \textcolor{green}{g}, 0\rangle}{\sqrt{2}},$$

$$|4\rangle = \frac{|0, \textcolor{red}{r}, \textcolor{green}{g}\rangle - |0, \textcolor{green}{g}, \textcolor{red}{r}\rangle}{\sqrt{2}},$$

$$|5\rangle = |0, d, 0\rangle,$$

$$|6\rangle = \frac{|\textcolor{red}{r}, d, \textcolor{green}{g}\rangle - |\textcolor{green}{g}, d, \textcolor{red}{r}\rangle}{\sqrt{2}},$$